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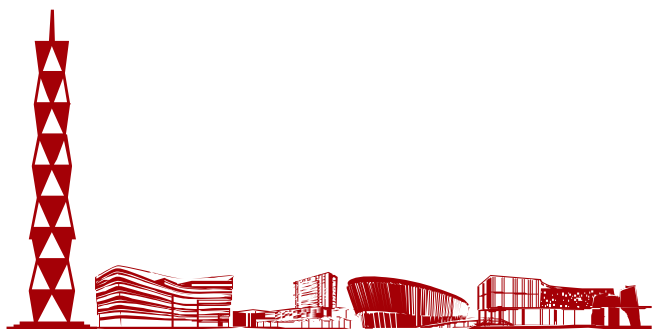
Investigating semantic expectation and predictive error in the visual cortex with a large fMRI vision-language dataset

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³*Zhejiang University, Hangzhou, China*

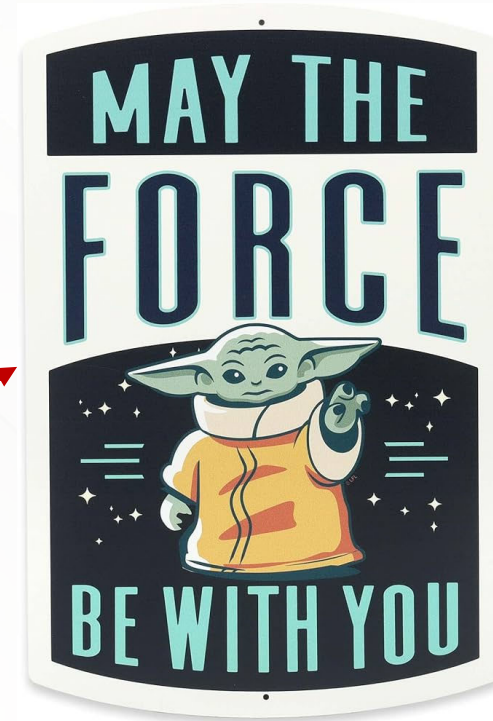


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Background



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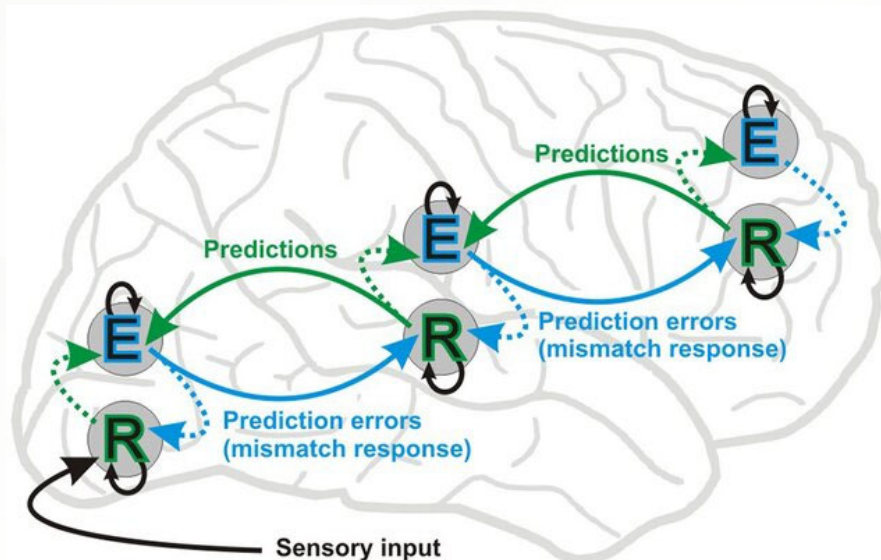


How does the brain extract and integrate semantic information across modalities?

Potential neural mechanism



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Simplified predictive coding model^[1]

E

Error coding unit

R

Representation unit



X

Too complex

Caption

Delayed-matching

Image

Simplified integration process

Two core questions:

1. Is semantic matching between caption and image based on the **predictive coding framework**?
2. **What** does neural response **encode** during cross-modal semantic matching?

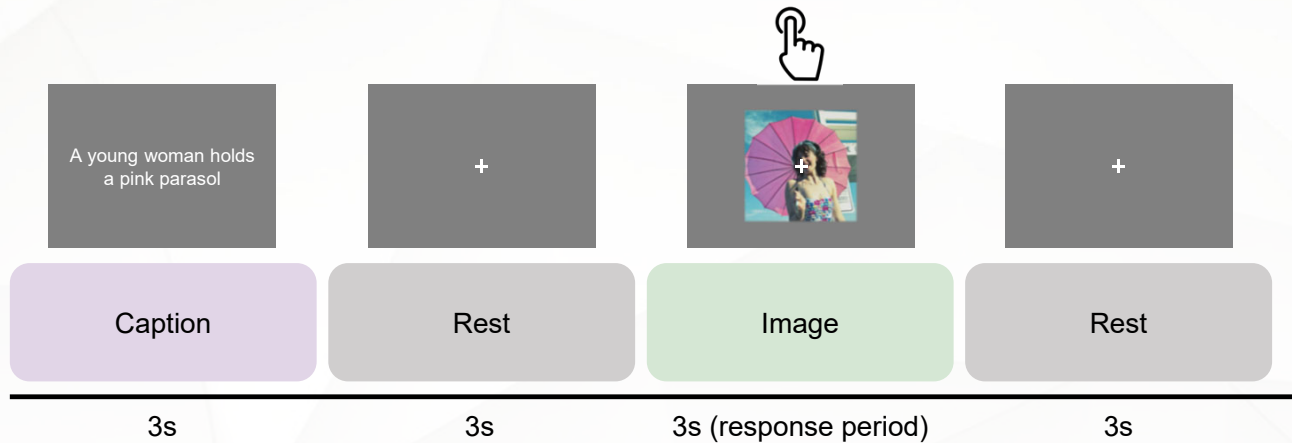
[1] Stefanics, Gábor, Jan Kremláček, and István Czigler. "Visual mismatch negativity: a predictive coding view." Frontiers in human neuroscience 8 (2014): 666.

[2] Marr, David. Vision: A computational investigation into the human representation and processing of visual information. MIT press, 2010.

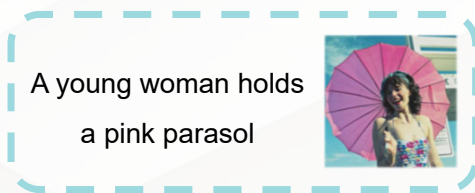
Experiment design & Dataset overview



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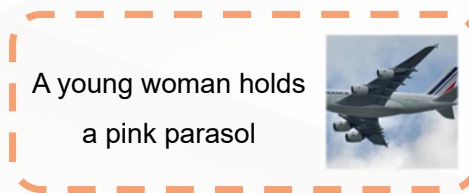


Behavior task: "Does the image match the caption?"



Behavior response: matched

Account for 90%



Behavior response: unmatched

Account for 10%

- 8 healthy subjects (mean age: 22.62)
 - 4 males and 4 females
- Over **200** hours of 3T MRI scanning
 - 4,400 caption-image pair trials
 - 4,000 matched trials & 400 unmatched trials
- Stimulus from COCO^[3] & COCO-CN^[4]

More information about dataset:

Abstract #**56.453**

A large-scale vision-language fMRI dataset for multi-modal semantic processing

Time: Tuesday, **May 20**, 2025, **2:45 – 6:45 pm**

Room: **Pavilion**

[3] Lin, Tsung-Yi, et al. "Microsoft coco: Common objects in context." Computer vision—ECCV 2014: 13th European conference, zurich, Switzerland, September 6-12, 2014, proceedings, part v 13. Springer International Publishing, 2014.

[4] Li, Xirong, et al. "COCO-CN for cross-lingual image tagging, captioning, and retrieval." IEEE Transactions on Multimedia 21.9 (2019): 2347-2360.

5

Question 1

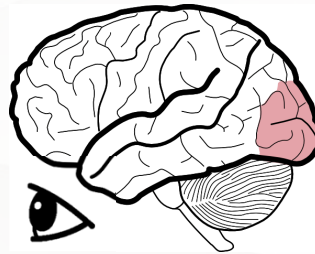


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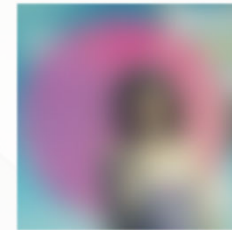
1. Is semantic matching between text and image based on predictive coding framework?

- Does cross-modal semantic expectation **exist** in the early visual cortex?

A young woman holds
a pink parasol



?

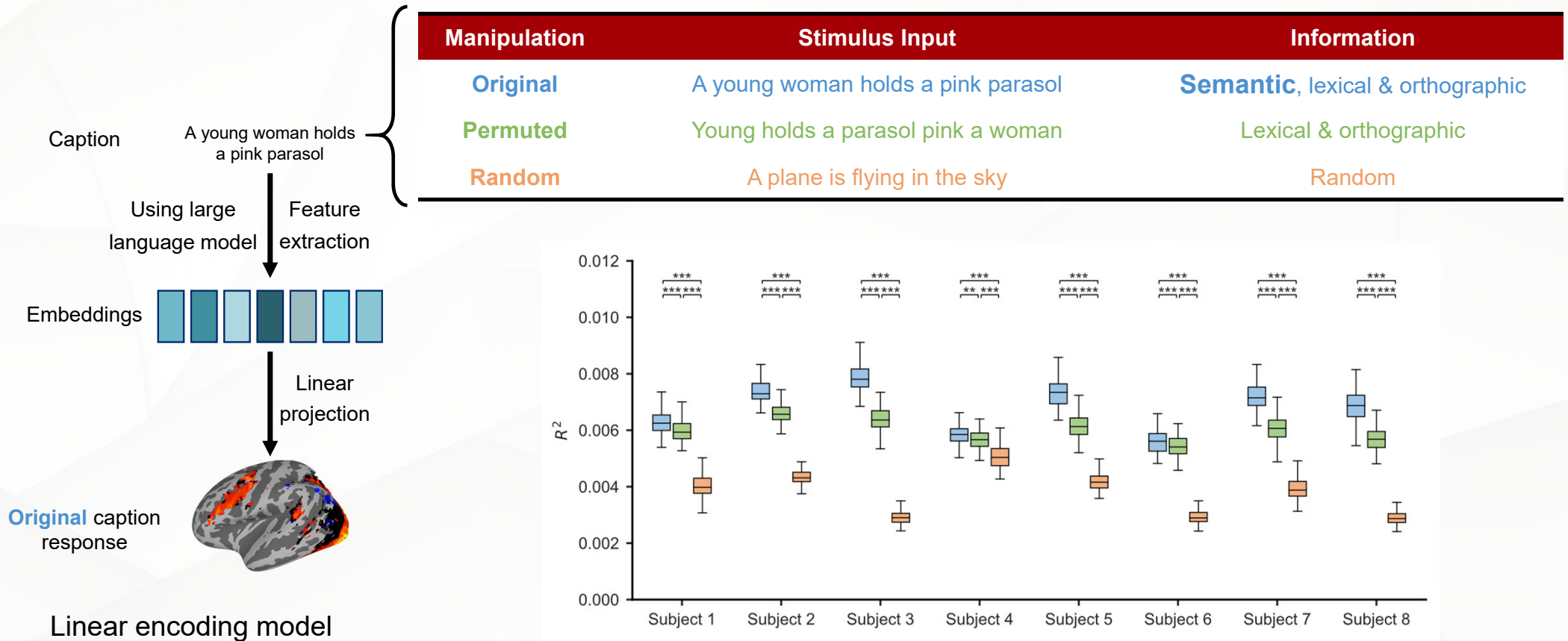


Expect semantics

Cross-modal semantic expectation exists in EVC



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EVC responses are significantly correlated with semantic expectation.

Question 1



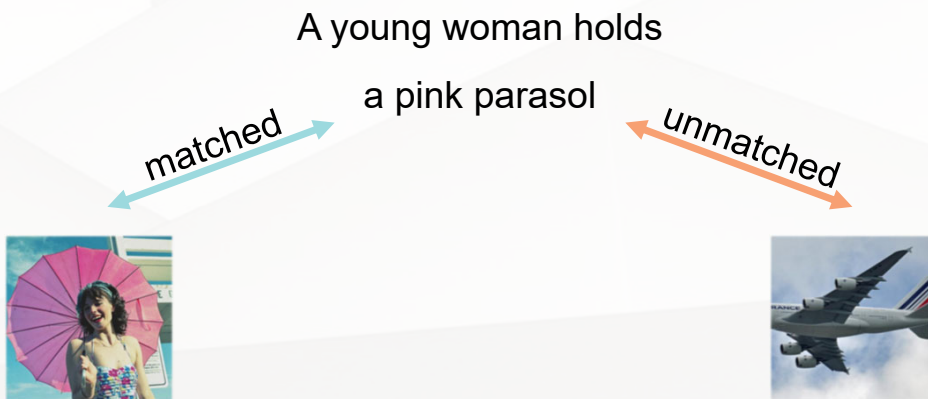
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1. Is semantic matching between text and image based on predictive coding framework?

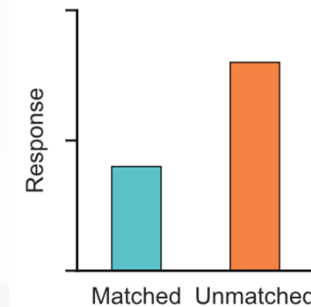
- Does cross-modal semantic expectation **exist** in the early visual cortex?

Yes, and **EVC** responses are significantly correlated with **semantic expectation**.

- Does cross-modal expectation suppression **occur** in this process? If so, **where** does it occur?



?



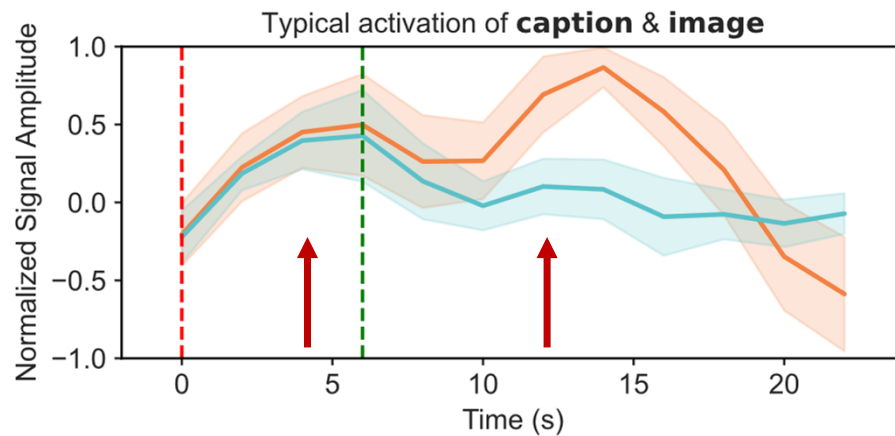
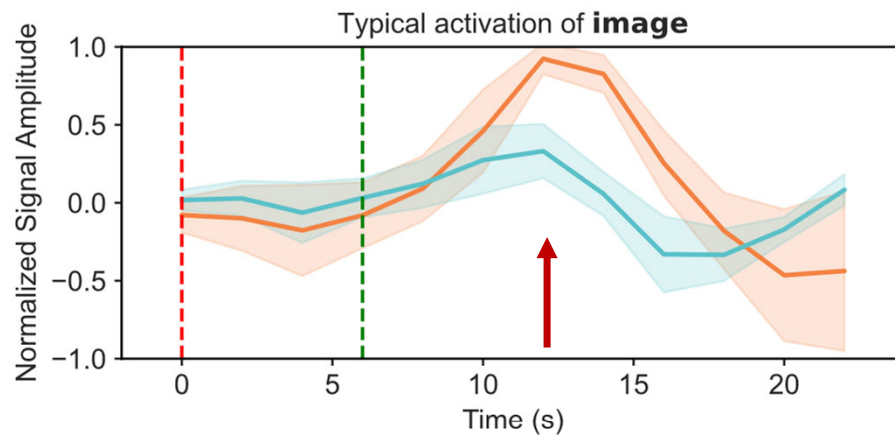
Hypothesis: Cross-modal expectation suppression

Expectation suppression is mainly along the ventral pathway

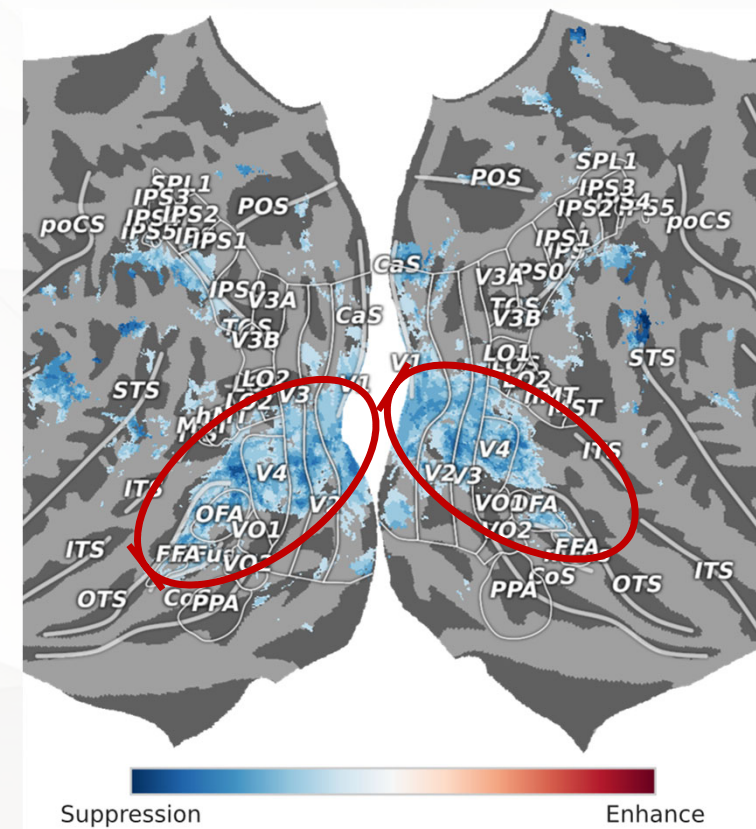


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— Matched — Unmatched - - - Caption onset - - - Image onset



Unmatched image response > Matched image response



Question 1



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1. Is semantic matching between text and image based on predictive coding framework?

- Does cross-modal semantic expectation **exist** in the early visual cortex?

Yes, and **EVC** responses are significantly correlated with **semantic expectation**.

- Does cross-modal expectation suppression **occur** in this process? If so, **where** does it occur?

Yes, and cross-modal expectation suppression is mainly along **the ventral pathway**.

Yes, **semantic matching** between text and image is based on **predictive coding** framework.

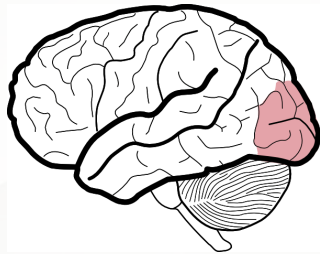
Question 2



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2. What does neural response encode during cross-modal semantic matching?

- **At which levels** are predictive errors encoded in the visual hierarchy?



? Encode low-level differences
(texture, shape, etc.)

? Encode high-level semantic
mismatches

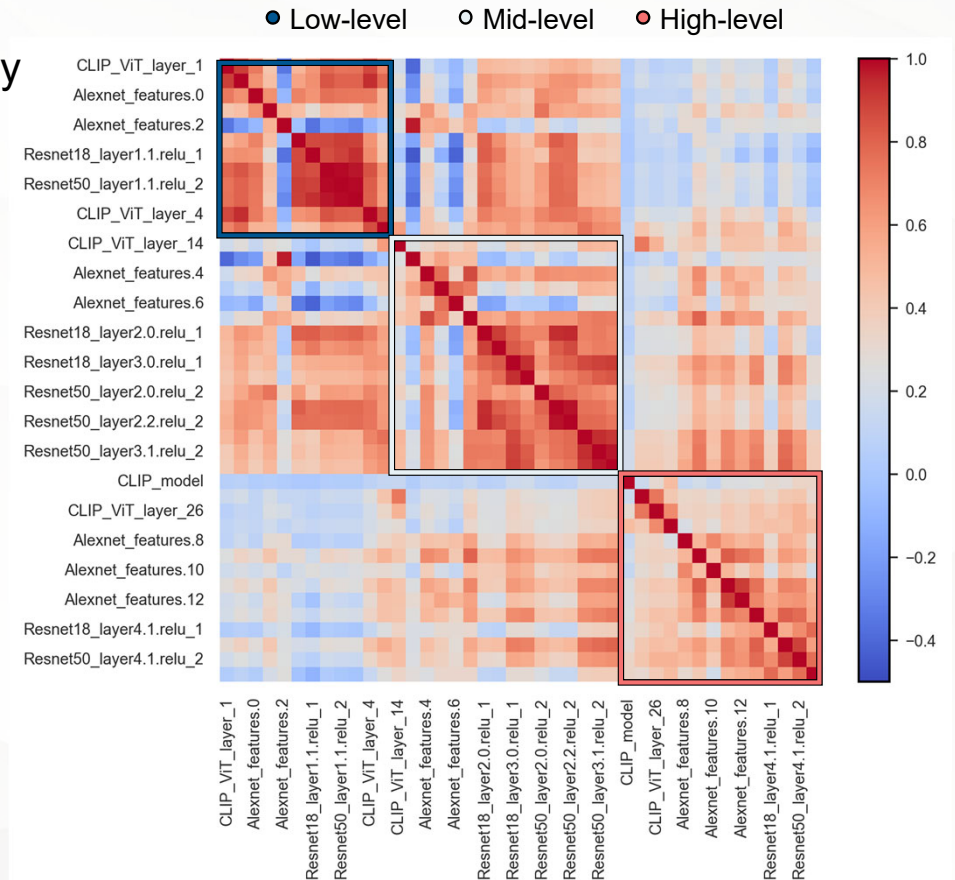
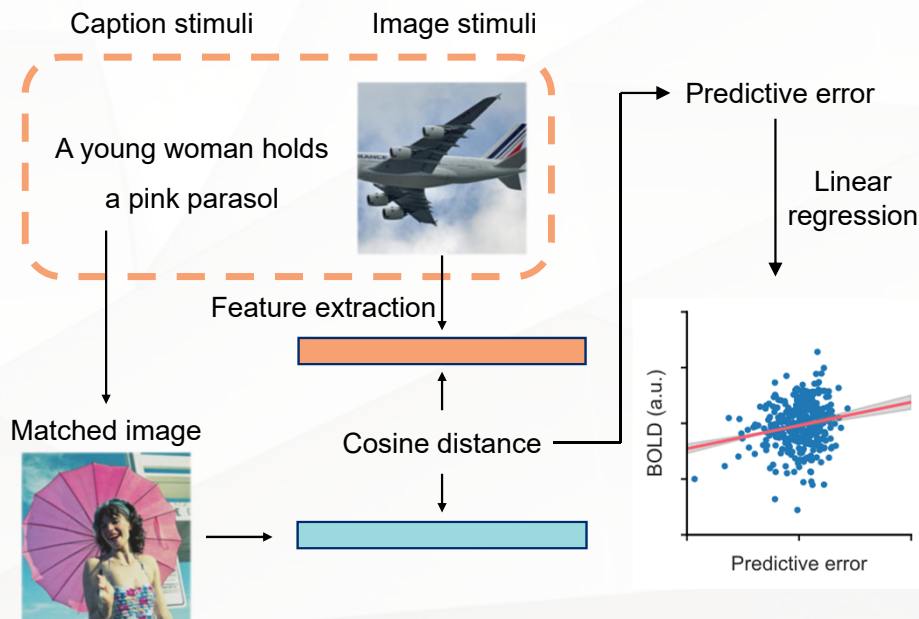
Feature dissimilarity as a proxy for predictive error



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Using **pretrained AI model feature** dissimilarity as a proxy

- Alexnet, Resnet 18, Resnet 50
- ViT, ViT from CLIP

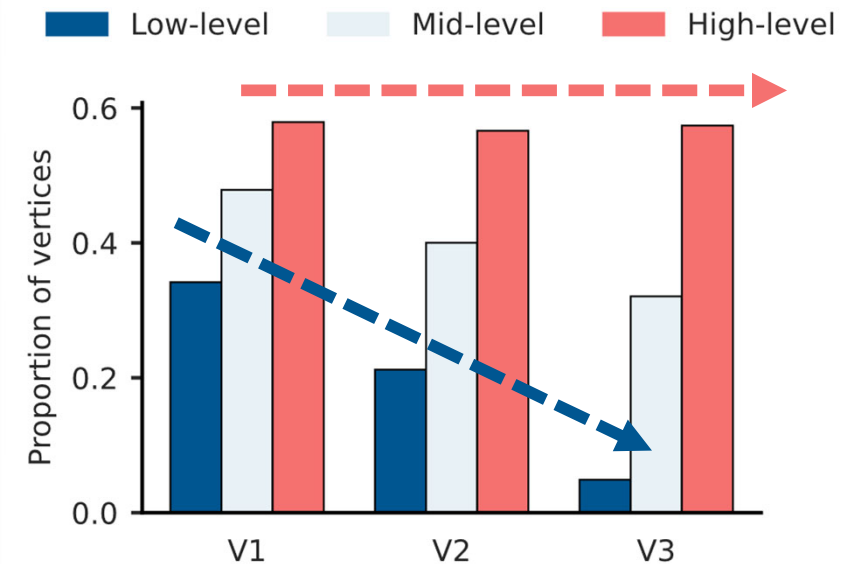
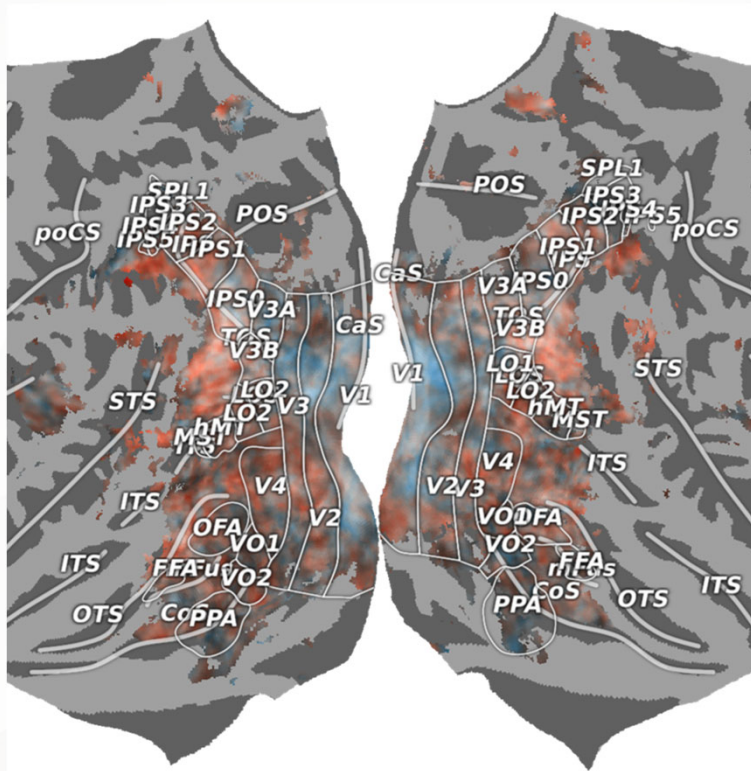


Correlation matrix of model features

The neural response encodes predictive errors



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- Low-level predictive errors decrease along the visual hierarchy.
- High-level predictive errors also can be represented in early visual region.

Question 2



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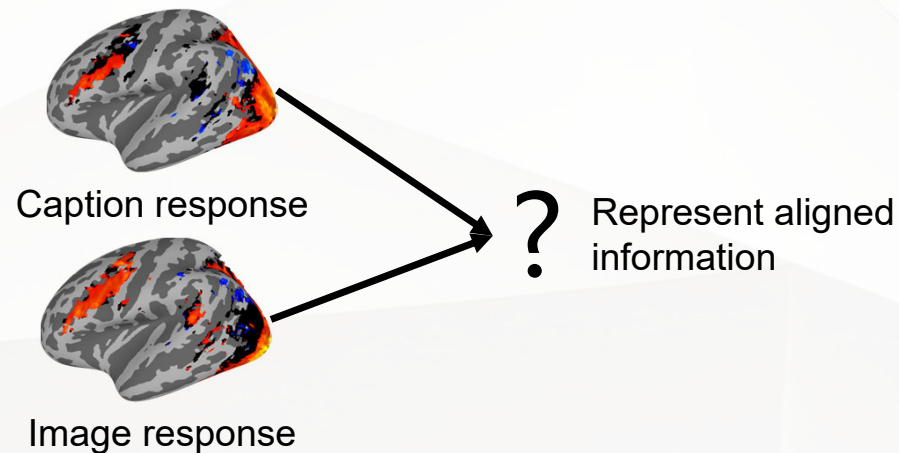
2. What does neural response encode during cross-modal semantic matching?

- **At which levels** are predictive errors encoded in the visual hierarchy?

Low-level predictive errors **decrease along the visual hierarchy**.

High-level predictive errors can be represented **in early visual regions**.

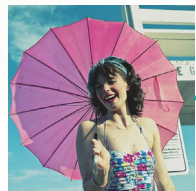
- Is **aligned** information **encoded** in EVC **across modalities**?



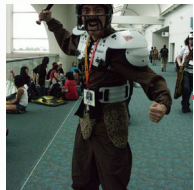
Modeling neural alignment across modality



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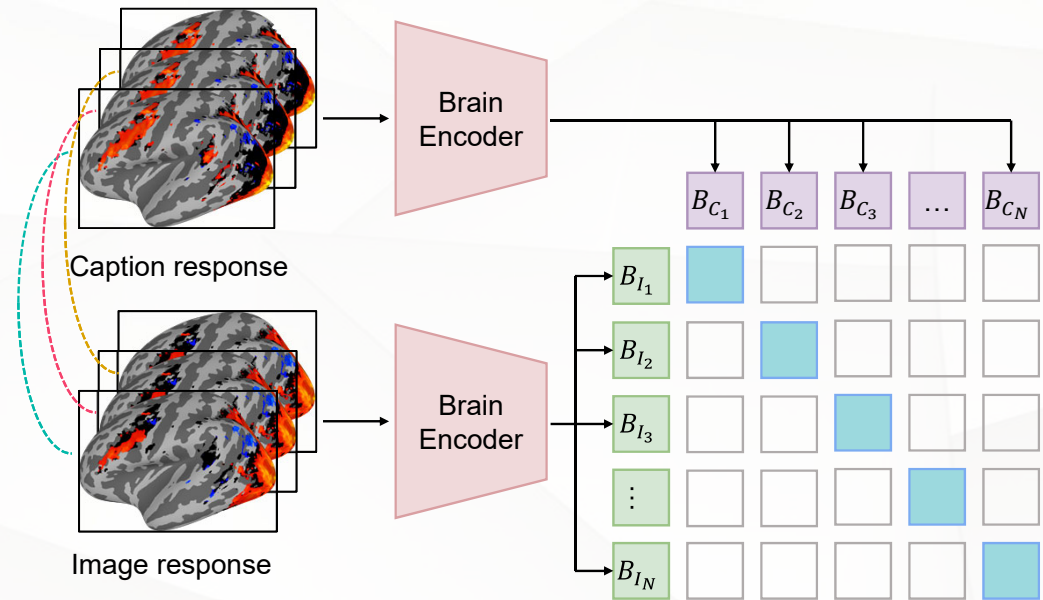
A young woman holding
a pink sun umbrella



A man is role-playing



A group of birds are
foraging in the wetland

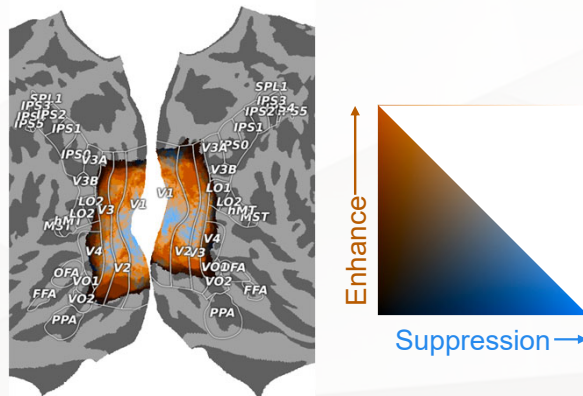
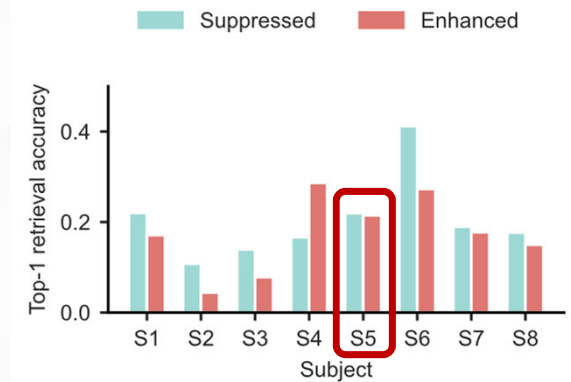
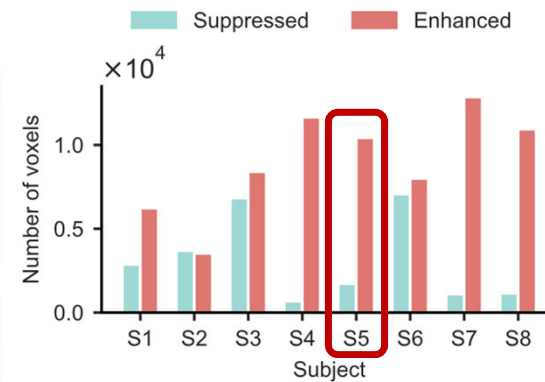
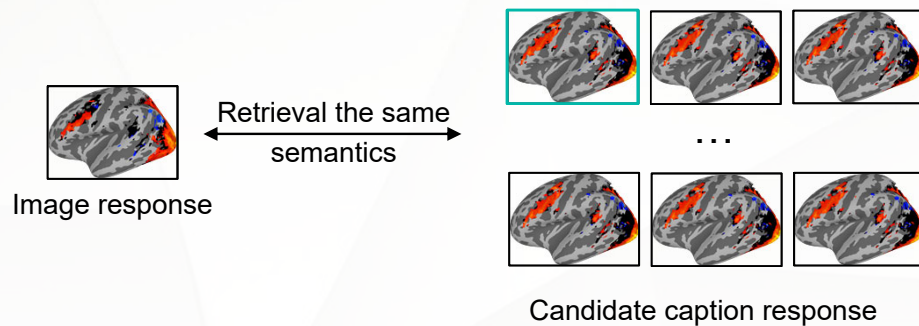


BrainCLIP to align brain response of caption & image

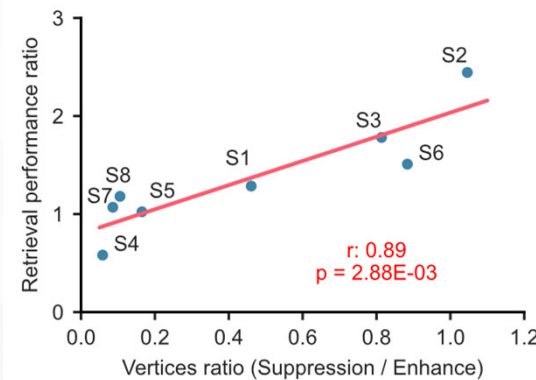
Aligned information is encoded in suppression area



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Enhance & Suppression area



A larger suppression area corresponds to higher retrieval performance.

Question 2



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2. What does neural response encode during cross-modal semantic matching?

- **At which levels** are predictive errors encoded in the visual hierarchy?

Low-level predictive errors **decrease along the visual hierarchy**.

High-level predictive errors can be represented **in low-level region**.

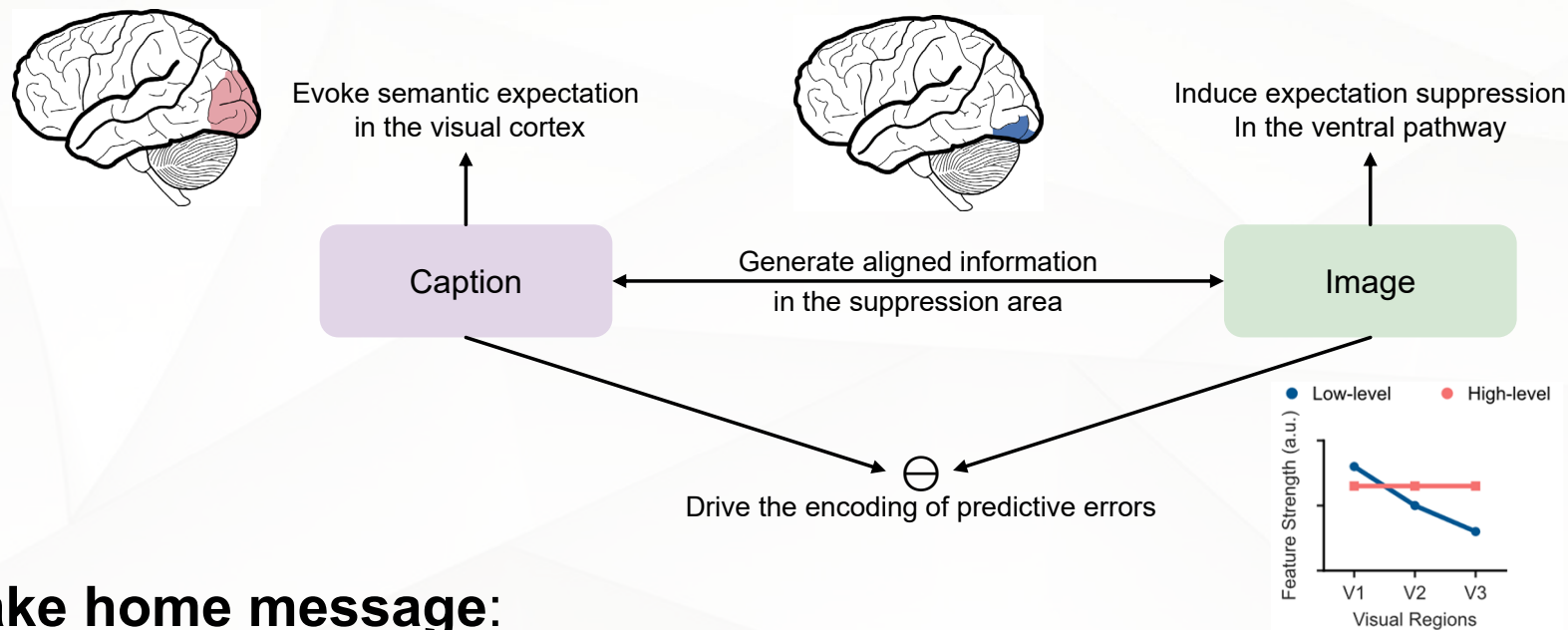
- Is **aligned** information **encoded** in EVC **across modalities**?

Yes — especially in **regions** showing expectation **suppression**.

Conclusion



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Take home message:

Early visual cortex **is not a passive feature detector** — this area can predict, compare and integrate semantic information across modalities.

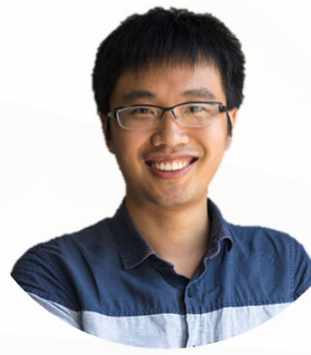
Acknowledgements



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Zhejiang University



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Shanghai Jiao Tong University

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Shiji Xiang (ShanghaiTech University)



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中华人民共和国科学技术部
Ministry of Science and Technology of the People's Republic of China



上海市科学技术委员会
(上海市外国专家局)



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Thanks!

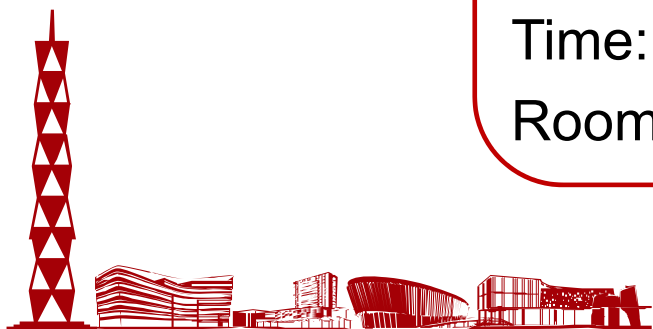
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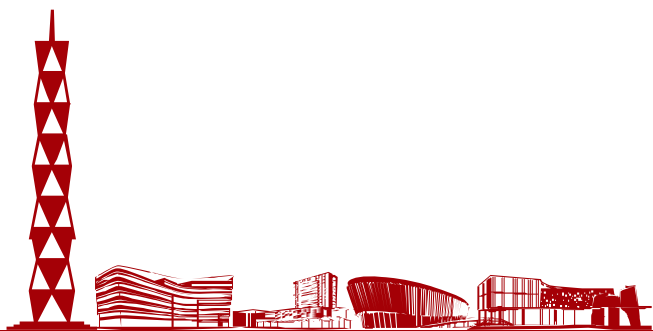
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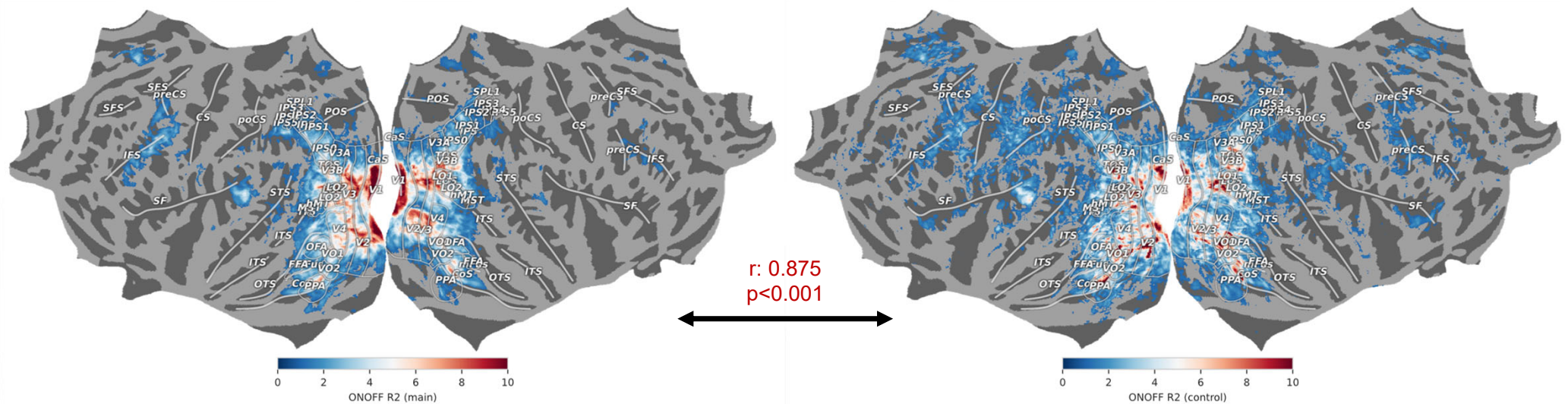
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Control experiment

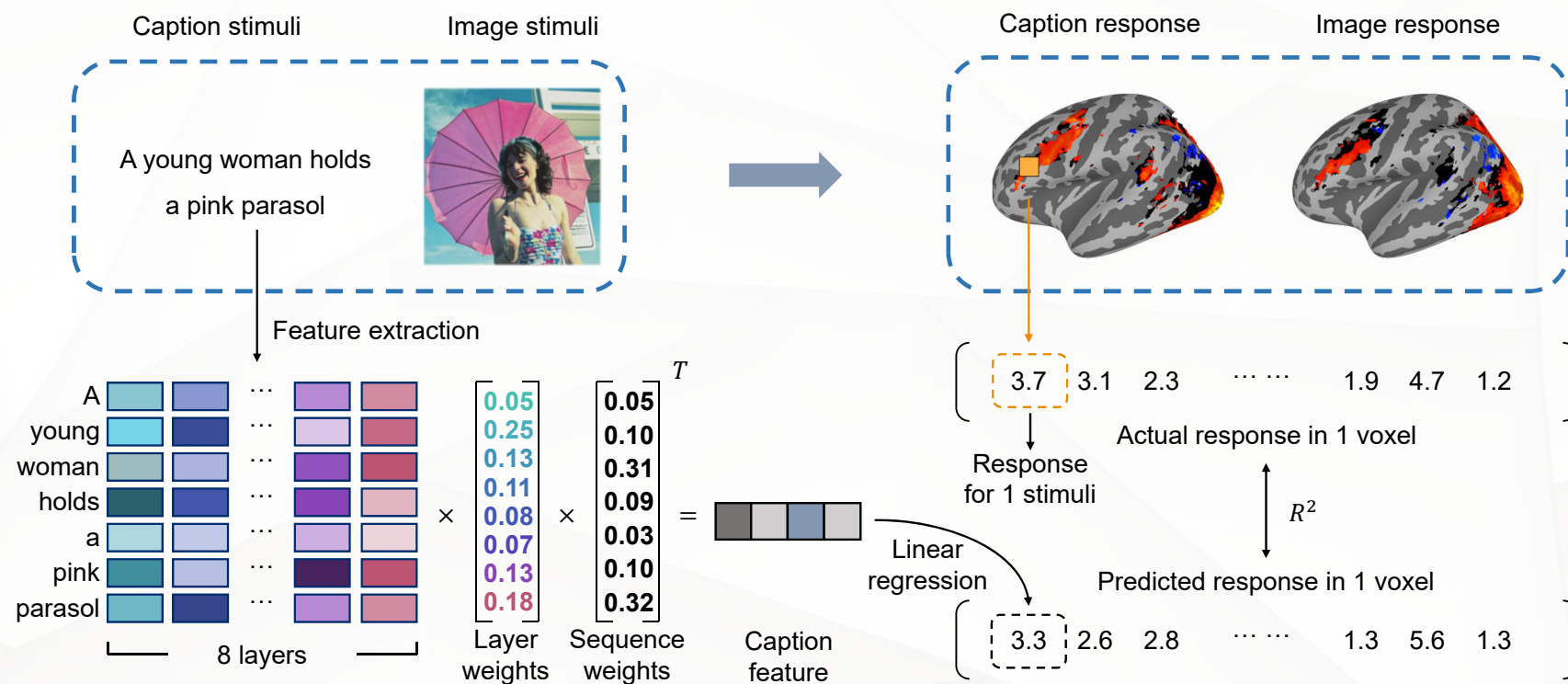


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	Matched: Unmatched	Runs	Trials
Main	9:1	200	4400
Control	1:9	16	400



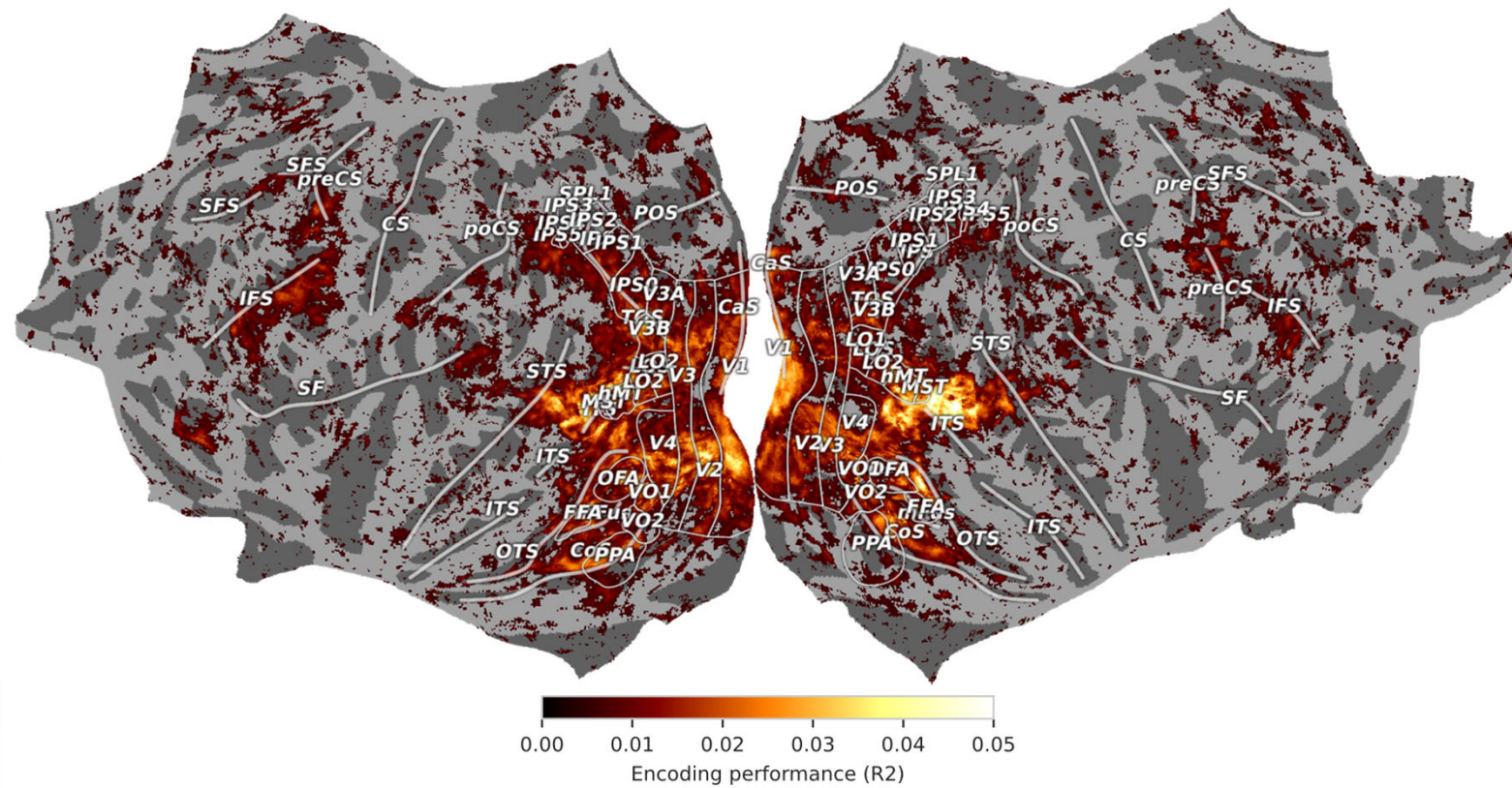
Architecture of encoding model



Encoding performance



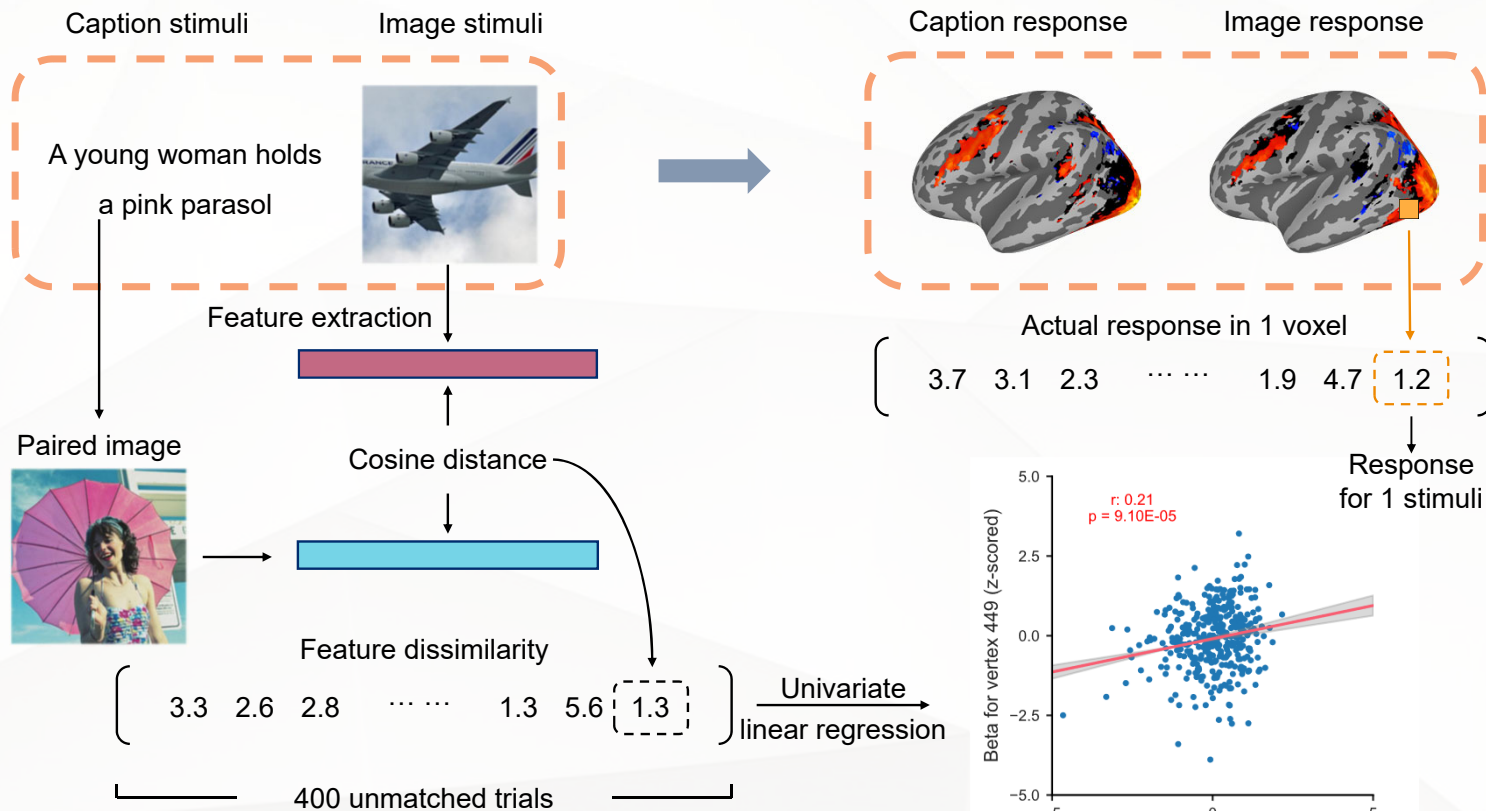
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Architecture of predictive error model



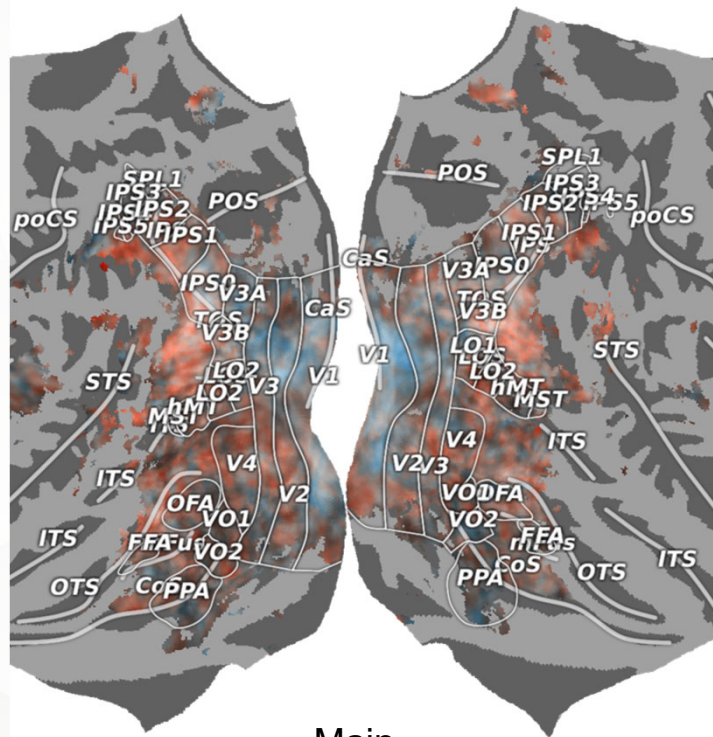
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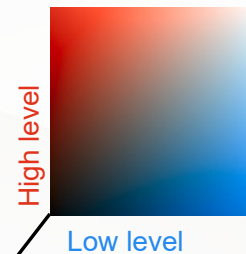
Level of predictive errors



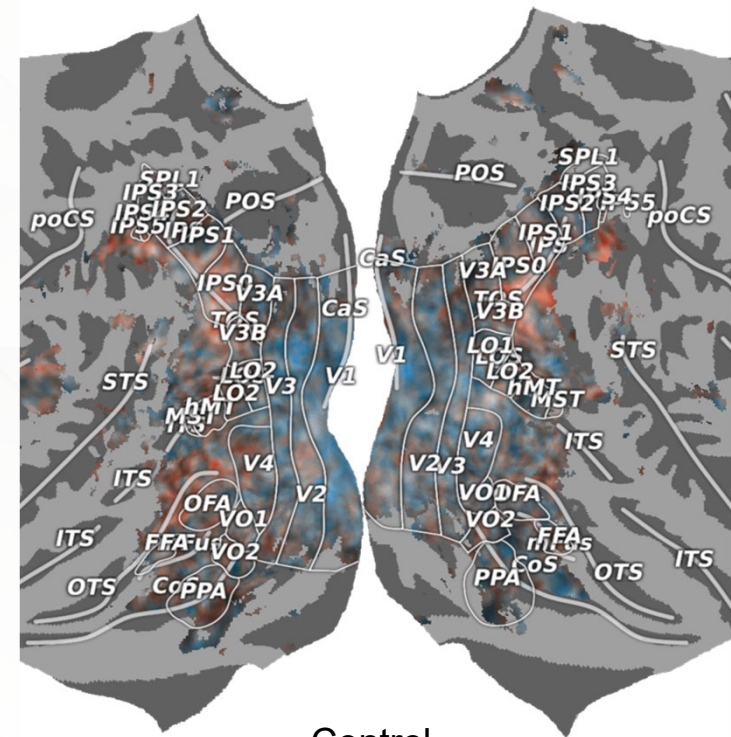
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Main



Not represent
predictive error

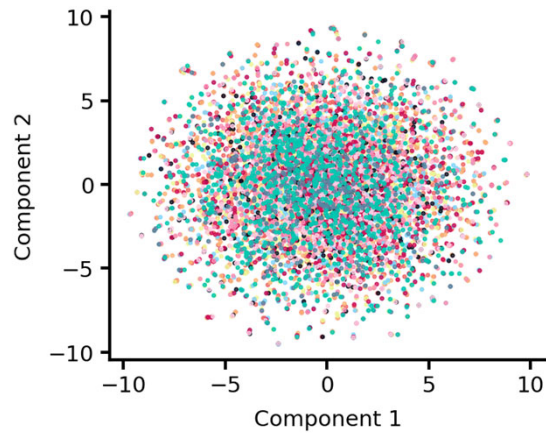


Control

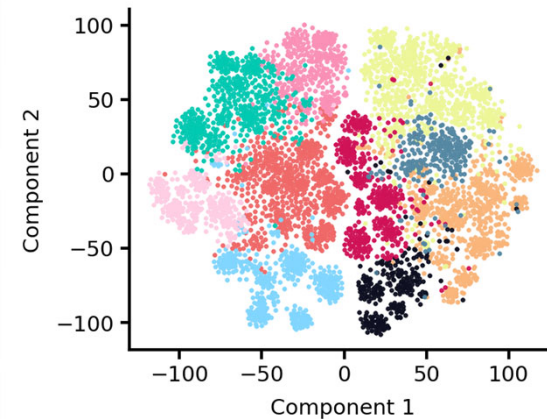
BrainCLIP can capture semantics across modalities



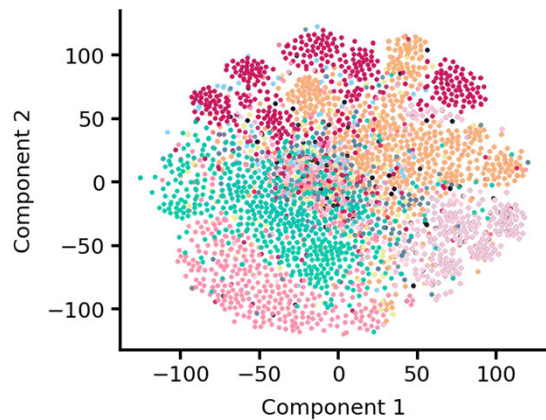
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TSNE map of raw brain semantic space



TSNE map of stimulus' CLIP feature space



TSNE map of encoded brain semantic space

